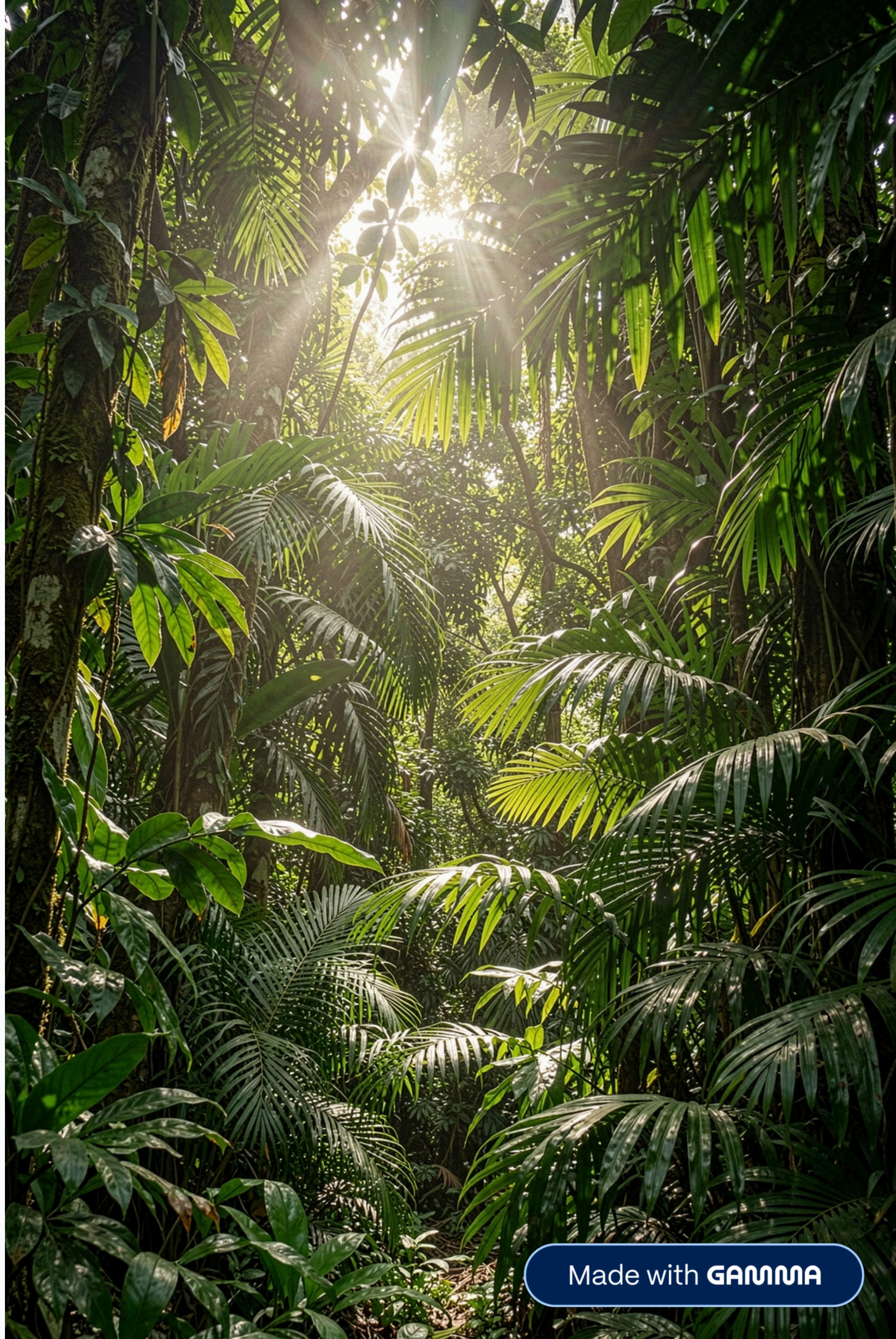
Ecology, Biodiversity & Conservation

A comprehensive study guide covering the foundational principles of ecology, population dynamics, community structure, and biodiversity conservation — with a focused lens on India and Assam's rich natural heritage. Designed for undergraduate students and competitive-exam candidates in ecology, conservation biology, and environmental science.

BASIC ECOLOGY | POPULATION & COMMUNITY | BIODIVERSITY & CONSERVATION



BASIC ECOLOGY

Principles of Ecology & the Ecosystem

What Is Ecology?

Ecology is the scientific study of interactions between organisms and their environment — both biotic (living) and abiotic (non-living). It operates at multiple levels: individual, population, community, ecosystem, and biosphere.

Core Principles

- Energy flows through ecosystems; matter cycles
- Organisms adapt to their environment over time
- Limiting factors control population size
- Ecological succession drives community change

The Ecosystem

An ecosystem is a functional unit of nature where living organisms interact with each other and their physical environment. Components include:

- **Producers** — autotrophs (plants, algae)
- **Consumers** — herbivores, carnivores, omnivores
- **Decomposers** — fungi, bacteria
- **Abiotic factors** — sunlight, water, soil, temperature

Ecosystems can be natural (forests, wetlands) or artificial (croplands, urban parks).

BASIC ECOLOGY

Food Chain, Food Web & Ecological Pyramids

Food Chain

A linear sequence showing energy transfer from producers to top consumers. Example: **Grass → Grasshopper → Frog → Snake → Eagle**. Each step is a **trophic level**; only ~10% of energy transfers upward (Lindeman's Law).

Food Web

A network of interconnected food chains showing multiple feeding relationships. More realistic than a single chain — if one species declines, alternatives buffer the ecosystem. Greater web complexity = greater stability.

Ecological Pyramids

Graphical representations of trophic structure: **Pyramid of Numbers** (individuals per level), **Pyramid of Biomass** (total mass), and **Pyramid of Energy** (always upright — energy decreases at each level). Inverted pyramids occur in aquatic biomass systems.

BASIC ECOLOGY

Energy Flow & Ecological Succession

Energy Flow in Ecosystems

Energy enters ecosystems as sunlight, is fixed by producers via photosynthesis, and flows unidirectionally through trophic levels. At each transfer, ~90% is lost as heat (Second Law of Thermodynamics). This is the **10% Rule** — explaining why food chains rarely exceed 4–5 levels.

> **Gross Primary Productivity (GPP)** = total energy fixed. **Net Primary Productivity (NPP)** = GPP minus respiration. NPP fuels all higher trophic levels.

Ecological Succession

The predictable, directional change in species composition over time in a given area.

- **Primary Succession** — begins on bare rock/lava; pioneer species (lichens, mosses) create soil
- **Secondary Succession** — follows disturbance (fire, logging); soil already present, faster recovery
- **Climax Community** — stable, self-perpetuating endpoint in equilibrium with climate

Pioneer

Lichens and moss colonize bare rock

Intermediate

Grasses and shrubs develop soil

Transitional

Young trees form a developing forest

Climax

Mature, stable forest community

Succession illustrates how ecosystems recover and mature over time, progressing from simple pioneer communities to complex, stable climax ecosystems.

POPULATION & COMMUNITY ECOLOGY

Population Ecology, Natality & Mortality

Population Ecology

Studies how populations change in size, density, and distribution over time. Key parameters: **natality** (birth rate), **mortality** (death rate), **immigration**, and **emigration**. Population growth follows either **J-shaped** (exponential) or **S-shaped** (logistic) curves.

Natality & Mortality

Natality = number of births per unit time; determines population increase. **Mortality** = number of deaths per unit time; reduces population. The balance between these, plus migration, defines the **population growth rate (r)**: **r = (B – D) + (I – E)**.

Carrying Capacity (K)

The maximum population size an environment can sustain indefinitely given available resources. As population approaches K, growth slows (density-dependent regulation). Exceeding K leads to population crash — a critical concept in wildlife management and conservation.

POPULATION & COMMUNITY ECOLOGY

Community Ecology & Population Estimation

Community Ecology

A **biological community** is all populations of different species living and interacting in a defined area. Key concepts:

- **Species richness** — number of species present
- **Species evenness** — relative abundance of each species
- **Keystone species** — disproportionately large ecological impact
- **Competitive exclusion** — two species cannot occupy identical niches
- **Mutualism, predation, parasitism** — interspecific interactions

Population Estimation Methods

- **Direct Count** — total census; used for large, visible species
- **Mark-Recapture (Lincoln Index)** — $N = (M \times C) / R$; ideal for mobile animals
- **Quadrat Sampling** — for plants and sessile organisms
- **Transect Method** — line or belt transects for density estimates
- **Indirect Signs** — pugmarks, scat, camera traps (used for tigers in India)

Biodiversity & Biodiversity Hotspots

What Is Biodiversity?

Biodiversity refers to the variety of life at all levels: **genetic diversity** (variation within species), **species diversity** (variety of species), and **ecosystem diversity** (variety of habitats). India is one of the world's **17 megadiverse countries**, hosting ~7–8% of all recorded species.

Biodiversity Hotspots

Regions with exceptional species richness and endemism under severe threat. India has **4 hotspots**:

- **Western Ghats** — endemic amphibians, mammals
- **Himalayas** — alpine flora, snow leopard habitat
- **Indo-Burma** — includes Northeast India (Assam)
- **Sundaland** — Nicobar Islands

Assam's Significance: Part of the Indo-Burma hotspot. Home to Kaziranga (one-horned rhino), Manas (UNESCO site), and rich avifauna. The Brahmaputra floodplains support unique wetland ecosystems.

BIODIVERSITY & CONSERVATION

Wildlife Conservation & Protected Areas in India & Assam

Kaziranga National Park, Assam

UNESCO World Heritage Site; home to **2/3 of the world's Greater One-Horned Rhinoceros**. Also supports tigers, elephants, and swamp deer. Established 1905; declared a National Park in 1974.

Manas National Park, Assam

UNESCO World Heritage Site and Tiger Reserve. Biosphere Reserve protecting rare species: **pygmy hog, hispid hare, Assam roofed turtle**. Located on the Bhutan-India border along the Manas River.

Protected Area Categories (India)

Under the **Wildlife Protection Act, 1972**: **National Parks** (strict protection, no human activity), **Wildlife Sanctuaries** (some human use permitted), **Tiger Reserves** (Project Tiger zones), **Biosphere Reserves** (core + buffer + transition zones), and **Conservation Reserves**.

BIODIVERSITY & CONSERVATION

Tiger Conservation, Ecotourism & Wildlife Management

Project Tiger & Tiger Conservation

Launched in **1973**, Project Tiger is India's flagship conservation program. India now has **53 Tiger Reserves** and ~3,167 tigers (2022 census) — over 75% of the world's wild tigers. Assam's reserves: **Kaziranga, Manas, Nameri, and Orang**. Conservation tools include camera trapping, M-STrIPES monitoring, and habitat corridors.

Ecotourism & Wildlife Management

Ecotourism promotes responsible travel to natural areas, supporting conservation and local livelihoods. In Assam, Kaziranga and Manas attract thousands of visitors annually, generating revenue for park management.

- **Habitat Analysis** — assesses suitability using GIS, remote sensing, field surveys
- **Wildlife Management** — population monitoring, anti-poaching, habitat restoration
- **Conservation Ethics** — intrinsic value of nature, intergenerational equity, precautionary principle

ADVANCED TOOLS & CONCEPTS

Ecological Perturbation, Remote Sensing & GIS

Ecological Perturbation

Any disturbance — natural (fires, floods, cyclones) or anthropogenic (deforestation, pollution, climate change) — that alters ecosystem structure and function. **Resilience** is an ecosystem's ability to recover; **resistance** is its ability to withstand change. Assam faces perturbations from flooding, encroachment, and infrastructure development.

Remote Sensing & GIS in Ecology

- **Satellite imagery** — monitors land-use change, deforestation, habitat fragmentation
- **NDVI** — measures vegetation health and density
- **GIS mapping** — overlays species distribution, protected areas, and threats
- **Applications in Assam** — flood mapping in Kaziranga, rhino habitat modeling, corridor planning

🌿 Key Takeaway

Ecology connects organisms to their environment; conservation applies that knowledge to protect biodiversity.

📡 Technology

Remote sensing and GIS are indispensable tools for modern conservation planning and monitoring.

🐯 India's Role

India leads global tiger conservation and hosts 4 biodiversity hotspots — a critical responsibility.